

- ⑤ Find the gradient of the tangent to the curve $y = 2x^2 - 3x - 7$ at the points where $x = 4, 2, 0, -2, -3$. Sketch the curve.
- ⑥ Find the equation of the tangent at the point $(2, 0)$ to the curve $y = x^2 - 2x$.
- ⑦ Find the point at which the tangent to the curve $y = 3x^2 - x$ at the point $(1, 2)$ cuts the x -axis.
- ⑧ Find the point at which the tangent to the curve $y = x^2 - 4x + 1$ at the point $(2, -3)$ cuts the y -axis.
- ⑨ Sketch the curve $y = 6 - 3x - x^2$, and calculate the gradient at the points where $x = 3, 1, -1, -3, -5$.
- ⑩ Find the coordinates of the point on the graph of $y = x^2 + 2x - 10$ at which the gradient is 8.
- ⑪ If $y = x^2 - 3x + 5$, find the coordinates of the point on the graph at which the tangent makes an angle of 45° with the positive direction of the x -axis.
- ⑫ Find the coordinates of the point on the graph of $y = 5x^2 + 8x - 1$ at which the gradient is -2 .
- ⑬ Find the coordinates of the points on the graph $y = \frac{8}{x}$ at which the gradient is -2 .
- ⑭ Calculate the gradients of the following curves at the points where x has the given values.
- $y = 5 - 3x^2$, at $x = 1\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -1$
 - $y = x + \frac{1}{x}$, at $x = 1, \frac{2}{3}, -\frac{1}{2}, -2$
 - $y = x^3 - 4x + 1$, at $x = 2, 0, -1, -3$
 - $y = x^3 - 5x^2 + 6x - 3$, at $x = 3, 1, -1, -2$
 - $y = 3 - \frac{1}{2x^2}$, at $x = 2, \frac{1}{2}, -\frac{2}{3}, -1$
- ⑮ If $y = (x - 1)(x^2 - 4)$, find the gradient of the curve at each of the points at which it cuts the x -axis. Hence sketch the curve.
- ⑯ Find the coordinates of the points on the graph of $y = 2x^3 + 3x^2 - 36x - 20$ at which the gradient is zero.

- ⑰ If $y = 2x^3 - 6x^2 - 15x + 19$, find the coordinates of the points on the graph at which the gradient is 3.
- ⑱ Find the coordinates of the points on the graph of $y = 3x^3 + 18x^2 + 24x + 5$ at which the gradient is -3 .
- ⑲ Find the equation of the tangent to the curve $y = x^2$ at the point (a, a^2) . Find the equations of the tangents to this curve which pass through the point $(1, -3)$.
- ⑳ The graph of $y = ax^2 + bx + c$ passes through the origin. The gradient of the tangent when $x = 1$ is 4 and when $x = 2$ is 5. Find the values of a, b and c .

Velocity and acceleration

Suppose that a particle's distance, s metres, after t seconds is given by $s = t^2 + 3t + 5$.

The velocity of the particle is the rate of change of s compared with t , i.e. $\frac{ds}{dt}$. Since $s = t^2 + 3t + 5$,

$$\text{then } \frac{ds}{dt} = 2t + 3.$$

Hence the velocity after t seconds is given by $2t + 3$.

Table 10.3 gives corresponding values for distance and time.

t	0	1	2	3	4	5
s	5	9	15	23	33	45

Table 10.3

The distance-time graph is shown in Figure 10.10.

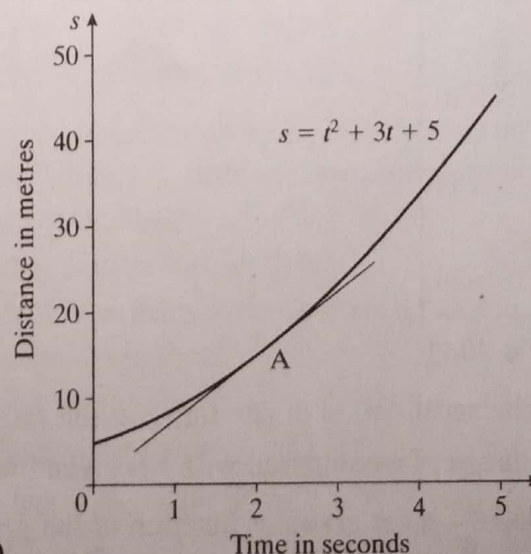


Fig. 10.10

The numerical value of the velocity at any point is given by $\frac{ds}{dt}$, i.e. by the gradient of the curve at that point.

$$\text{If } t = 0, \frac{ds}{dt} = 2 \times 0 \times 3 = 3$$

The initial velocity is therefore 3 m/s.

$$\text{If } t = 2, \frac{ds}{dt} = 2 \times 2 \times 3 = 7$$

Hence the velocity after 2 seconds is 7 m/s.

Acceleration is the rate of change of velocity compared with time. Hence if the velocity is v m/s after t seconds, then the acceleration is given by $\frac{dv}{dt}$.

When distance is measured in metres and time in seconds, velocity is expressed in metres per second (m/s) and acceleration in metres per second per second (m/s^2).

Suppose a particle is moving such that its velocity in terms of t is given by $v = 6t - t^2$.

To draw the velocity-time graph, make a table of values as in Table 10.4.

t	0	1	2	3	4	5	6	7	8
v	0	5	8	9	8	5	0	-7	-16

Table 10.4

The velocity-time graph is shown in Fig. 10.11.

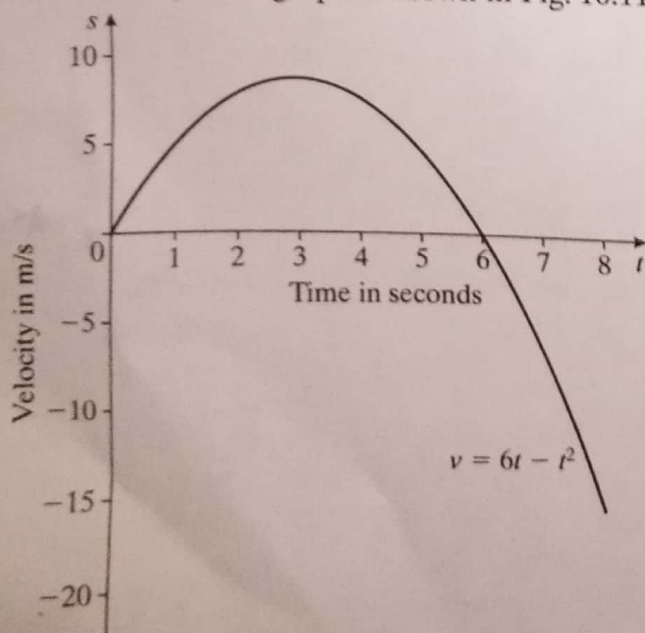


Fig. 10.11

The acceleration at any time, i.e. the rate of change of v compared with t , is given by $\frac{dv}{dt}$, and $\frac{dv}{dt}$ is the gradient function of the graph.

Therefore, acceleration is given by the gradient of the velocity-time graph, just as velocity is given by the gradient of the distance-time graph.

In Fig. 10.11, when $t = 3$ the gradient of the curve is zero.

Hence, after 3 seconds the acceleration is zero. This may be checked by differentiation:

$$v = 6t - t^2$$

$$\Rightarrow \frac{dv}{dt} = 6 - 2t$$

When $t = 3$,

$$\frac{dv}{dt} = 6 - 2 \times 3 = 0$$

For values of t from 0 to 3, the gradient is positive,

i.e. the acceleration is positive,

i.e. the velocity is increasing.

When $t = 3$, v reaches a maximum of 9, and the acceleration is momentarily zero. The velocity then begins to decrease, since the gradient, i.e. the acceleration, is negative.

When $t = 6$, the curve cuts the horizontal axis, i.e. after 6 seconds the velocity is zero. For higher values of t the velocity is negative, which means that the particle is going to the opposite direction.

Example 16

After t seconds a particle has travelled a distance s metres, where $s = -27t + 15t^2 - t^3$. When is its velocity zero? When is its acceleration zero? Find its velocity and acceleration after:

- a 2 seconds b 6 seconds c 10 seconds.

$$s = -27t + 15t^2 - t^3$$

$$\Rightarrow \frac{ds}{dt} = -27 + 30t - 3t^2$$

$$\text{i.e. velocity } v = -27 + 30t - 3t^2$$

$$\Rightarrow \frac{dv}{dt} = 30 - 6t$$

$$\text{i.e. acceleration} = 30 - 6t$$

If the velocity is zero, then

$$-27 + 30t - 3t^2 = 0$$

$$\Leftrightarrow -3(t^2 - 10t + 9) = 0$$

$$\Leftrightarrow (t - 1)(t - 9) = 0$$

$$\Leftrightarrow t = 1 \text{ or } 9$$

Hence, the velocity is zero after 1 second and after 9 seconds.

If the acceleration is zero, then

$$30 - 6t = 0$$

$$\Leftrightarrow t = 5$$

The acceleration is zero after 5 seconds.

a When $t = 2$:

$$\begin{aligned}\text{velocity in m/s} &= -27 + 30 \times 2 - 3 \times 2^2 \\ &= -27 + 60 - 12 \\ &= 21\end{aligned}$$

$$\begin{aligned}\text{and acceleration in m/s}^2 &= 30 - 6 \times 2 \\ &= 18\end{aligned}$$

b When $t = 6$:

$$\begin{aligned}\text{velocity in m/s} &= -27 + 30 \times 6 - 3 \times 6^2 \\ &= -27 + 180 - 108 \\ &= 45\end{aligned}$$

$$\begin{aligned}\text{and acceleration in m/s}^2 &= 30 - 6 \times 6 \\ &= -6\end{aligned}$$

c When $t = 10$:

$$\begin{aligned}\text{velocity in m/s} &= -27 + 30 \times 10 - 3 \times 10^2 \\ &= -27 + 300 - 300 \\ &= -27\end{aligned}$$

$$\begin{aligned}\text{and acceleration in m/s}^2 &= 30 - 6 \times 10 \\ &= -30\end{aligned}$$

Therefore:

after 2 s, vel. is 21 m/s and accel. is 18 m/s²

after 6 s, vel. is 45 m/s and accel. is -6 m/s²

after 10 s, vel. is -27 m/s and accel. is -30 m/s².

Exercise 10f

- ① A particle moves along a straight line in such a way that after t seconds it has gone s metres, where $s = t^2 + t$. Draw the distance-time graph for values of t from 0 to 4, and by measuring the gradient find the velocity of the particle:

a after 1 second b after 3 seconds.

Check your results by differentiation.

- ② A particle moves along a straight line in such a way that after t seconds its velocity is v m/s, where $v = t^2 - t + 2$. Draw the velocity-time graph for values of t from 0 to 4, and by measuring the gradient find the acceleration of the particle after:

a 2 seconds b 3 seconds c $\frac{1}{2}$ second.

Check your results by differentiation.

- ③ After t seconds the velocity of a moving particle is v m/s, where $v = 2t + 1$. By drawing the velocity-time graph, show that the acceleration is constant, and find its value. Check by differentiation.
- ④ After t seconds a moving body has gone s metres, where $s = 4t^2 - 4t + 1$. Calculate the velocity after 2 seconds and after 3 seconds. When is the body momentarily at rest?
- ⑤ After t seconds a moving body has a velocity of v m/s, where $v = 5t^2 - 12t + 7$. Calculate the acceleration after 2 seconds and after 3 seconds.
- ⑥ A moving body goes s metres in t seconds, where $s = 4t^2 - 3t + 5$. Find its velocity after 4 seconds. Show that the acceleration is constant, and find its value.
- ⑦ A particle travels s metres in t seconds, where $s = t^2 - 2t + 5$. Show by differentiation that the acceleration is constant, and find its value. Calculate the velocity after 1 second, 3 seconds and 4 seconds, and check by drawing the distance-time graph for values of t from 0 to 5.
- ⑧ A stone is thrown vertically into the air, and its height is s metres after t seconds where $s = 29.4t - 4.9t^2$.
- a After how many seconds does it reach its greatest height?
- b What is its greatest height?
- c What is its initial velocity?
- ⑨ A ball is projected up a sloping groove, and after t seconds its distance from the starting point is s cm, where $s = 24t - 3t^2$.
- a What is its initial velocity?
- b After how many seconds does it reach its furthest point?
- c What is its velocity after 8 seconds?
- d What is the value of s if $t = 8$?
- e What is its acceleration?

- 10 After t seconds a particle has gone s metres, where $s = t^3 - 6t^2 + 9t + 5$.

After how many seconds is its velocity zero?

When is its acceleration zero?

Find its velocity and acceleration:

- a initially b after 4 seconds.

- 11 A particle moves in such a way that after t seconds it has gone s metres, where $s = 5t + 15t^2 - t^3$.

Find its velocity and acceleration after:

- a 3 seconds b 6 seconds.

c When is the acceleration zero?

- 12 The distance s metres that a particle has gone in t seconds is given by $s = 20t + 12t^2 - 2t^3$.

After how many seconds is its acceleration zero?

Find its velocity and acceleration after:

- a 1 second b 4 seconds.

Turning points

Maxima and minima

Fig. 10.12 represents the path of a ball kicked from A, reaching its greatest height BD, and striking the ground at C. Between A and B, when the ball is climbing, the gradient is positive, but steadily decreases to zero at B. Past B the ball is descending, and the path has a negative gradient.

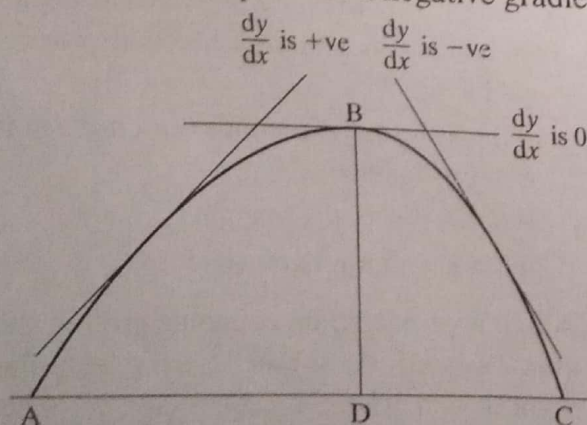


Fig. 10.12

A point such as B in Fig. 10.12 is called a **turning point**. The gradient at the turning point of any curve is always zero.

Since the height of the ball is *greatest* at B, point B is said to be a **maximum point**. Hence point B is a **maximum turning point**.

Fig 10.13 is a sketch of the curve $y = x^2 + 2$.

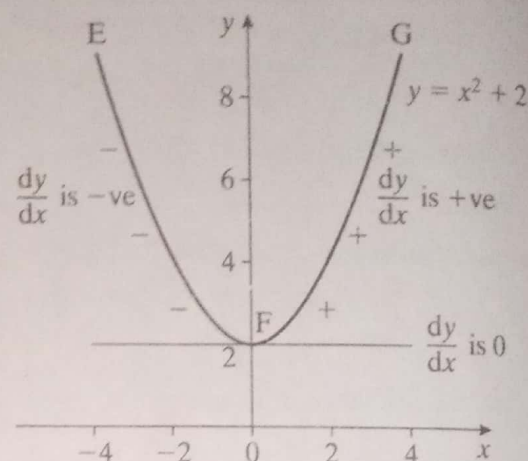


Fig. 10.13

The gradient of the curve at F is zero. F is therefore a turning point of the curve $y = x^2 + 2$.

Following the curve from E to G, the gradient is negative between E and F, but increases steadily to zero at F. Past F, the gradient is positive, increasing further to G.

The value of y at F is 2. This is the *least* value that y can have. Point F is called a **minimum point**. Hence point F is a **minimum turning point**.

The general form of a quadratic equation is $y = ax^2 + bx + c$. When a is positive, the curve is like a valley, such as PQR in Fig. 10.14a in which y has a least value QS. Q is therefore a minimum turning point of the curve PQR.

When a is negative, the curve is like a mountain top, such as UVW in Fig. 10.14b in which y has a greatest value VX. V is therefore a maximum turning point of the curve UVW.

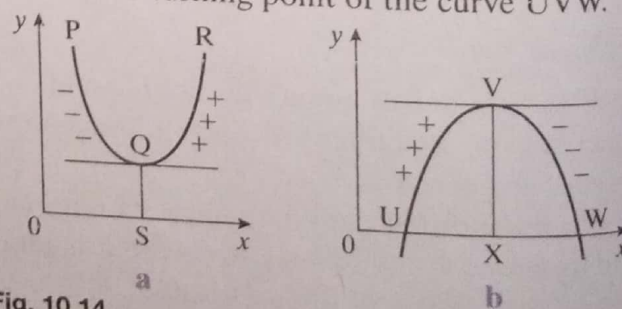


Fig. 10.14

Notice, from Fig. 10.14, that on passing along the curves from left to right:

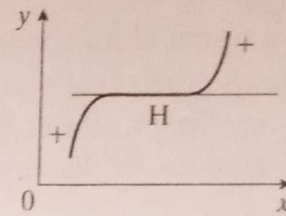
- 1 When a point is a maximum turning point, the curve has the greatest value, and its gradient is zero at that point. The gradient changes from *positive* to *negative*.
- 2 When a point is a minimum turning point, the curve has the least value, and its gradient is zero at that point. The gradient changes from *negative* to *positive*.

Point of inflexion

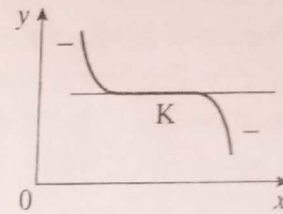
Fig. 10.15 represents sections of two curves with turning points at H and K.

The gradient is zero at both H and K. However, on each side of H the gradient is positive. Similarly, the gradient is negative on both sides of K. A point such as H or K is called a **point of inflexion**.

Table 10.5 summarises the distinction between the turning points. The symbols $-$ and $+$ are used to denote whether the values of the gradients on either side of the turning points are negative or positive.



a



b

Fig. 10.15

	left value	turning point at $x = a$	right value	diagrammatic representation	turning point
$\frac{dy}{dx}$	+	0	-		maximum
$\frac{dy}{dx}$	-	0	+		minimum
$\frac{dy}{dx}$	+	0	+		inflexion
$\frac{dy}{dx}$	-	0	-		inflexion

Table 10.5

Example 17

Find the maximum or minimum value of y on the curve $y = 4x - x^2$.

$$y = 4x - x^2$$

$$\Rightarrow \frac{dy}{dx} = 4 - 2x$$

The gradient is zero when

$$4 - 2x = 0$$

$$\Leftrightarrow 2(2 - x) = 0$$

$$\Rightarrow x = 2$$

and $y = 4 \times 2 - 2^2 = 4$

The turning point is $(2, 4)$.

Considering the sign of the gradient on the left and right of $(2, 4)$:

Just to the right of $(2, 4)$, x is just greater than 2, and $\frac{dy}{dx}$ is negative. Thus the shape of the curve at $(2, 4)$ is as sketched in Fig. 10.16.

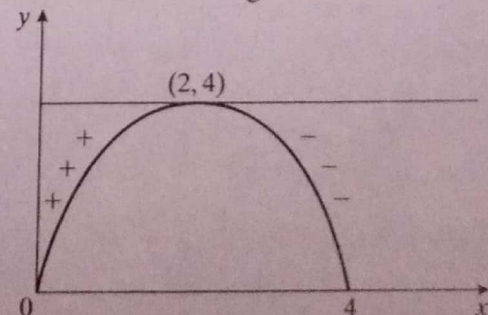


Fig. 10.16

The maximum value of y is $+4$.

Curve sketching

When drawing a *sketch* of a curve, you should show all the characteristic features of the curve. Accurate plotting of points is less important. Information concerning the turning points is of major importance.

Example 18

Find the maximum and minimum values of the function $2x^3 - 3x^2 + 12x + 8$. Hence sketch the curve whose equation is $y = 2x^3 - 3x^2 - 12x + 8$.

$$\begin{aligned} \text{If } y &= 2x^3 - 3x^2 - 12x + 8 \\ \text{then } \frac{dy}{dx} &= 6x^2 - 6x - 12 \end{aligned}$$

At a turning point, $\frac{dy}{dx} = 0$

$$\begin{aligned} \Leftrightarrow 6(x^2 - x - 2) &= 0 \\ \Leftrightarrow 6(x - 2)(x + 1) &= 0 \\ \Leftrightarrow x &= 2 \text{ or } -1. \end{aligned}$$

Hence the turning points occur where $x = 2$ or -1 .

The next section shows how to distinguish between them.

a If x is a little less than 2,

$$\begin{aligned} x - 2 \text{ is negative and } x + 1 \text{ is positive} \\ \Rightarrow 6(x - 2)(x + 1) \text{ is negative.} \end{aligned}$$

If x is a little greater than 2,

$$\begin{aligned} x - 2 \text{ is positive and } x + 1 \text{ is positive} \\ \Rightarrow 6(x - 2)(x + 1) \text{ is positive.} \end{aligned}$$

As x passes through the value 2, $\frac{dy}{dx}$ changes from negative to positive. Therefore a minimum occurs when $x = 2$. This minimum value of $2x^3 - 3x^2 - 12x + 8$ is:

$$\begin{aligned} 2 \times 2^3 - 3 \times 2^2 - 12 \times 2 + 8 \\ = 16 - 12 - 24 + 8 \\ = -12 \end{aligned}$$

b If x is a little less than -1 (e.g. -1.1),

$$\begin{aligned} x - 2 \text{ is negative and } x + 1 \text{ is negative} \\ \Rightarrow 6(x - 2)(x + 1) \text{ is positive.} \end{aligned}$$

If x is a little greater than -1 (e.g. -0.9),

$$\begin{aligned} x - 2 \text{ is negative and } x + 1 \text{ is positive} \\ \Rightarrow 6(x - 2)(x + 1) \text{ is negative.} \end{aligned}$$

As x passes through the value -1 , $\frac{dy}{dx}$ changes from positive to negative. Therefore a maximum occurs when $x = -1$.

This maximum value of $2x^3 - 3x^2 - 12x + 8$ is:

$$\begin{aligned} 2 \times (-1)^3 - 3 \times (-1)^2 - 12 \times (-1) + 8 \\ = -2 - 3 + 12 + 8 \\ = 15 \end{aligned}$$

The characteristic features of the curve are now known. In addition, find the point at which the curve cuts the y -axis.

When $x = 0$,

$$\begin{aligned} y &= 2 \times 0^3 - 3 \times 0^2 - 12 \times 0 + 8 \\ &= 8 \end{aligned}$$

The curve can now be sketched as in Fig. 10.17. Notice that graph paper is not needed for a sketch.

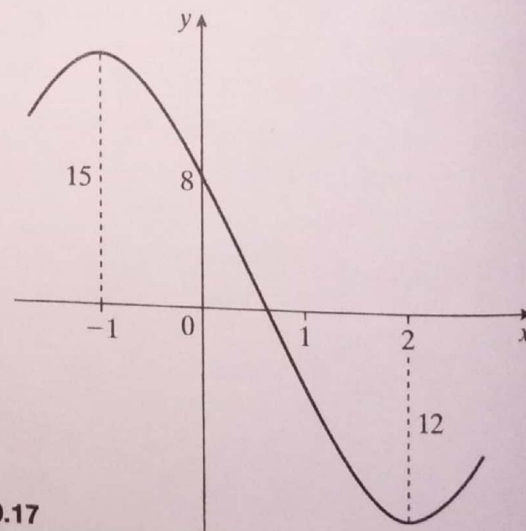


Fig. 10.17

Example 19

Investigate the turning points, if any, of $x^3 - 3x^2 + 3x + 2$, and sketch the graph of the function.

$$\text{Let } y = x^3 - 3x^2 + 3x + 2$$

$$\text{then } \frac{dy}{dx} = 3x^2 - 6x + 3$$

At a turning point $\frac{dy}{dx} = 0$,

$$\Rightarrow 3(x^2 - 2x + 1) = 0$$

$$\Leftrightarrow 3(x - 1)^2 = 0$$

$$\Rightarrow x = 1 \text{ (twice)}$$

The gradient is zero only at the point where $x = 1$.

If x is a little less than 1, $3(x - 1)^2$ is positive.
 If x is a little greater than 1, $3(x - 1)^2$ is positive.
 Therefore the gradient is positive on both sides of the point at which it is zero. Hence, there is a point of inflexion where $x = 1$, and there is no maximum or minimum turning point.

$$\text{If } x = 1, \\ y = 1^3 - 3 \times 1^2 + 3 \times 1 + 2 = 3$$

The point of inflexion is at (1, 3).

$$\text{When } x = 0, \\ y = 0^3 - 3 \times 0^2 + 3 \times 0 + 2 = 2$$

The curve cuts the y -axis at (0, 2).

The curve can now be sketched as in Fig. 10.18.

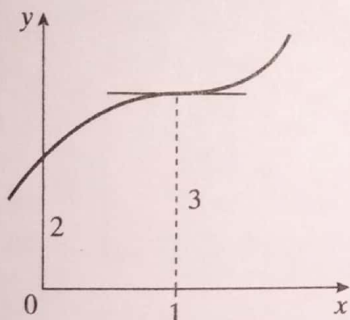


Fig. 10.18

Exercise 10g

For the following functions, find the values of x at which turning points occur, and distinguish between them. Find the value of the function at each turning point, and sketch the graph.

- | | |
|---------------------------|---------------------------|
| ① $x^2 - 6x + 3$ | ② $7 - 4x - x^2$ |
| ③ $6x - x^2$ | ④ $x^2 + 3x$ |
| ⑤ $2x^2 - x^3$ | ⑥ $x^3 - 3x - 1$ |
| ⑦ $3x^2 + 2x - 5$ | ⑧ $4 - 12x - 3x^2$ |
| ⑨ $x(5 - 2x)$ | ⑩ $x^3 - 12x + 2$ |
| ⑪ $2x^3 - 9x^2 + 12x - 4$ | ⑫ $1 + 9x + 3x^2 - x^3$ |
| ⑬ $1 - 3x - 3x^2 - x^3$ | ⑭ $x^3 - 6x^2 + 12x - 11$ |
| ⑮ $x + \frac{1}{x} + 2$ | ⑯ $2x^3 - 3x^2 - 36x + 6$ |
| ⑰ $6 + 12x - 3x^2 - 2x^3$ | ⑱ $3 - x^3$ |
| ⑲ $2x^3 + 3x^2 - 36x - 6$ | ⑳ $x^3 + 2$ |
| ㉑ $x^2 + \frac{2}{x}$ | ㉒ $2x + \frac{8}{x} + 3$ |
| ㉓ $x^3 - 5x^2 + 7x - 3$ | ㉔ $29 - 27x + 9x^2 - x^3$ |
| ㉕ $x^2 - \frac{16}{x}$ | |

Real-life problems

The previous examples involved finding the turning points of a given function. In the problems which follow, a variable is chosen and the function of this variable has to be constructed.

Notice that the constructed function must be in terms of *only one* variable, so that if two letters are originally used, one must be expressed in terms of the other.

Example 20

A farmer has 120 metres of fencing with which to enclose a rectangular field, using a wall for one side. Find the maximum area that he can enclose.

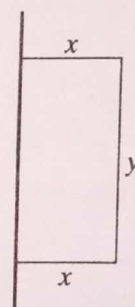


Fig. 10.19

Let the side parallel to the wall be of length y metres, and the other two sides x metres each.

$$\text{Then } 2x + y = 120$$

$$\Leftrightarrow y = 120 - 2x$$

If the area of the field is A square metres,

$$\begin{aligned} A &= xy \\ &= x(120 - 2x) \\ &= 120x - 2x^2 \end{aligned}$$

$$\therefore \frac{dA}{dx} = 120 - 4x$$

For a maximum or minimum,

$$120 - 4x = 0$$

$$\Leftrightarrow x = 30$$

If x is slightly < 30 , $120 - 4x$ is positive.

If x is slightly > 30 , $120 - 4x$ is negative.

In passing through the value 0, $\frac{dA}{dx}$ changes from +ve to -ve. Therefore $x = 30$ gives a maximum.

$$\text{If } x = 30, y = 120 - 2x = 120 - 60 = 60.$$

The field of maximum area measures 30 m by 60 m.

Its area is 1 800 square metres.

Example 21

A window is in the shape of a rectangle surmounted by a semicircle, the diameter of the semicircle being equal to the width of the rectangle. Find the radius of the semicircle if the window is to admit as much light as possible, given that the perimeter is 20 metres.

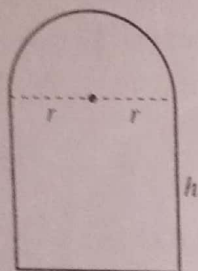


Fig. 10.20

Let r m and h m be as in Fig. 10.20, and let the area of the window be A m².

$$\text{Then } \pi r + 2r + 2h = 20$$

$$\Leftrightarrow 2h = 20 - \pi r - 2r$$

$$A = \frac{1}{2}\pi r^2 + 2rh$$

$$= \frac{1}{2}\pi r^2 + r[20 - \pi r - 2r]$$

$$= 20r - \frac{1}{2}\pi r^2 - 2r^2$$

$$\therefore \frac{dA}{dr} = 20 - \pi r - 4r$$

For a max. or min.:

$$20 - r(\pi + 4) = 0$$

$$\Leftrightarrow r = \frac{20}{\pi + 4}$$

If r is slightly $< \frac{20}{\pi + 4}$, $20 - r(\pi + 4)$ is +ve.

If r is slightly $> \frac{20}{\pi + 4}$, $20 - r(\pi + 4)$ is -ve.

In passing through the value 0, $\frac{dA}{dr}$ changes from + to -.

So $r = \frac{20}{\pi + 4}$ gives a maximum.

The area of the window is a maximum when the radius of the semicircle is $\frac{20}{\pi + 4}$ metres.

Exercise 10h

- ① A rectangular sheet of metal measures 8 cm by 3 cm. From each corner a square of side x cm is removed, and the flaps so formed are bent up to make a small open box. Show that the volume of the box is $(4x^3 - 22x^2 + 24x)$ cm³, and find its maximum value.

- ② In question 1, if the width of the metal sheet has been 5 cm instead of 3 cm, what would have been the maximum volume of the box?
- ③ If $A = xy$ where $3x + y = 18$, find the maximum value of A .
- ④ If $P = 9x + 8y$ where $xy = 2$, find the minimum value of P , taking x and y to be both positive.
- ⑤ Find the maximum area of a rectangle whose perimeter is 32 metres.
- ⑥ A piece of wire 36 cm long is cut and bent to make the shape shown in Fig. 10.21. Find the dimensions that will give the maximum volume outlined by the wire.

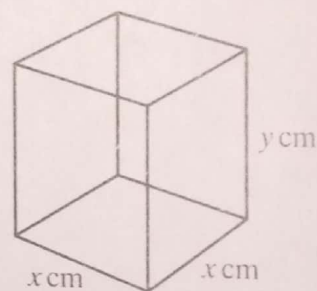


Fig. 10.21

- ⑦ A water-tank is open at the top, on a square base. If the volume of the tank is to be 32 m³, find the least area of metal sheet that can be used.
- ⑧ When an object is projected into the air its height, s metres, after t seconds is given by the formula $s = 98t - 4.9t^2$. Find the greatest height to which it rises.
- ⑨ A rectangular sheet of metal measures 20 cm by 15 cm. Equal squares of side x cm are cut from two adjacent corners, and the flaps so formed are bent up to form the tray of a shovel, as in Fig. 10.22. Find the value of x if the volume of the shovel is a maximum.

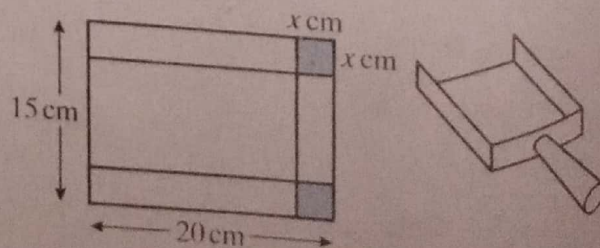


Fig. 10.22

- 10 Fig. 10.23 shows the shape to which some wire is bent to reinforce a package. If the volume of the package is to be a maximum, and the total length of the wire is 60 cm, find the length marked y cm.

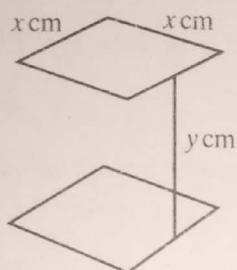


Fig. 10.23

- 11 A sheet of metal is 24 cm square. From each corner equal squares are removed, and the flaps so formed are bent up to make an open box. Find its maximum volume.
- 12 A farmer uses some 2 m hurdles to enclose a rectangular field of area 200 m^2 . If one side of the field is a river bank and needs no fencing, find the least number of hurdles needed.
- 13 A swimming pool is of the shape shown in Fig. 10.24. Find the maximum area of the pool if the perimeter is 132 m.

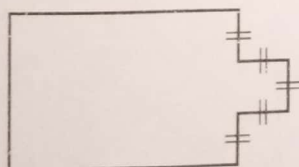


Fig. 10.24

- 14 Sheet metal is used for making a closed rectangular box with square ends. If the total area of the metal is 216 cm^2 , find the dimensions that will give the greatest volume.
- 15 A swimming pool is of the shape shown in Fig. 10.25, the radius of each semicircular part being $\frac{1}{4}$ of the width. If the perimeter of the pool is 200 m, find the width when the area is a maximum. (Leave π in the answer.)

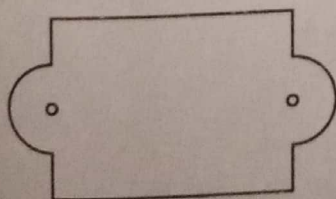


Fig. 10.25

- 16 Find the maximum volume of a solid circular cylinder if its total surface area is $6\pi \text{ cm}^2$. (Leave π in the answer.)
- 17 Find the minimum surface area of a solid circular cylinder if its volume is $16\pi \text{ cm}^3$. (Leave π in the answer.)
- 18 A piece of wire of given length is bent to form the perimeter of a sector of a circle. If the area of the sector is a maximum, find the angle between the bounding radii. (Leave π in the answer.)

SUMMARY

- 1 A function $f(x)$ tends towards a limit, k , if $|f(x) - k| \rightarrow 0$ as $x \rightarrow \text{some value } a$.
- 2 The *gradient function* of a curve is used to find the gradient of a given curve at a given point.
- 3 $\frac{dy}{dx}$ is the *differential coefficient* (or *derived function* or *derivative*) of y with respect to x . It measures *the rate of change* of y compared to x .
- 4 If $y = Ax^n$, then $\frac{dy}{dx} = Anx^{n-1}$;
if $y = x^{-n}$, then $\frac{dy}{dx} = -nx^{-n-1}$;
if $y = k$, then $\frac{dy}{dx} = 0$.
- 5 If y is a function of t and t is a function of x , then $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$ (the *chain rule*).
- 6 Given u and v are functions of x :
• if $y = uv$, then $\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$ (the *product rule*)
• if $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ (the *quotient rule*).
- 7 Differential calculus may be applied to solving problems involving gradients, velocity and acceleration, turning points, curve sketching, and may be applied to real-life problems. See Examples 13–21.

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Determine the equation of a line, or curve, given its gradient function
- 2 Use the inverse of differentiation to find the general solutions of simple differential equations
- 3 Apply the rules of integration to integrate algebraic expressions
- 4 Calculate the arbitrary constant, given sufficient relevant information
- 5 Apply integral calculus to solving problems relating to
 - coordinate geometry
 - displacement, velocity, acceleration
- 6 Evaluate definite integrals in the form $\int_a^b f(x)dx$
- 7 Find the area under a curve by integrating between limits.

Teaching and learning materials

Teacher: Charts showing standard integrals; calculus-related computer instructional materials where available

11.1 Reverse of differentiation

This chapter, which is closely related to Chapter 10, is mainly concerned with using a given expression for $\frac{dy}{dx}$ to find y in terms of x .

Therefore it may be thought of as the inverse or **reverse of differentiation**.

It is helpful to begin with a review of gradient functions, covered in the previous chapter.

Gradient functions (revision)

Example 1

- Find the gradient functions of *i* $y = 4x$, *ii* $y = 4x + 2$, *iii* $y = 4x + c$, where c is a constant.
- Sketch the equations of the lines in part *a*.

The gradient functions are:

$$\text{i} \quad \frac{dy}{dx} = \frac{d}{dx}(4x) = 4$$

$$\text{ii} \quad \frac{dy}{dx} = \frac{d}{dx}(4x + 2) = 4$$

$$\text{iii} \quad \frac{dy}{dx} = \frac{d}{dx}(4x + c) = 4$$

Hence the gradient functions are the same and are all equal to 4.

- The three equations represent parallel straight lines with gradient 4 as shown in Fig. 11.1.

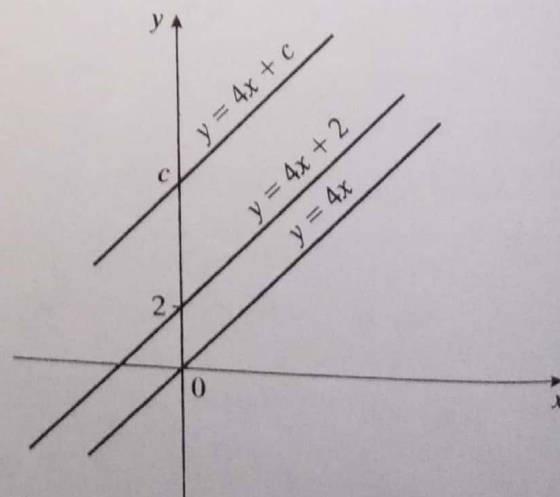


Fig. 11.1

Suppose the equation of a gradient function is given as $\frac{dy}{dx} = \frac{1}{2}$.

The constant gradient $\frac{1}{2}$ indicates a straight line $y = \frac{1}{2}x$, a straight line of gradient $\frac{1}{2}$ through the origin. Differentiating $y = \frac{1}{2}x$ leads to $\frac{dy}{dx} = \frac{1}{2}$.

However, as seen in Example 1, this is not the only possibility. Any straight line of gradient $\frac{1}{2}$ may be written as $y = \frac{1}{2}x + c$, where c is a constant. This is the most general equation which gives $\frac{dy}{dx} = \frac{1}{2}$.

Hence the equation $\frac{dy}{dx} = \frac{1}{2}$ is the gradient function for the equation $y = \frac{1}{2}x + c$, that is, *all straight lines of gradient $\frac{1}{2}$* , some of which are shown in Fig. 11.2.

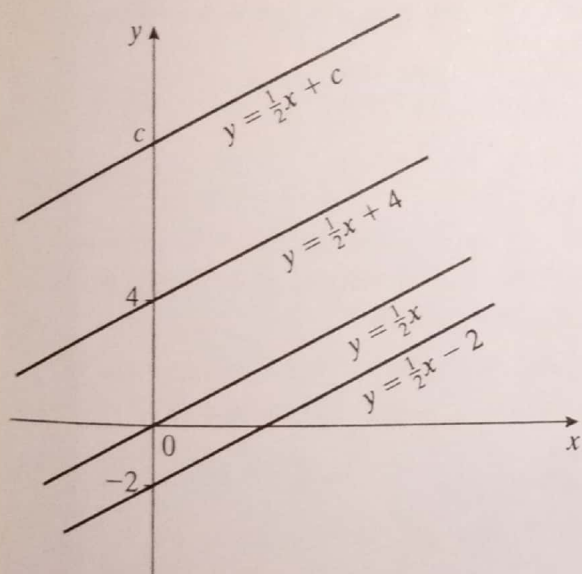


Fig. 11.2

Gradient functions are not always a constant.

For example, if $\frac{dy}{dx} = 2x$, $y = x^2$ is a curve with this gradient function. However, $y = x^2 + c$, where c is a constant is the most general equation leading to $\frac{dy}{dx} = 2x$ on differentiation.

Hence, $\frac{dy}{dx} = 2x$ is the gradient function of the equation $y = x^2 + c$, that is, *a family of curves 'parallel' to $y = x^2$* as shown in Fig. 11.3

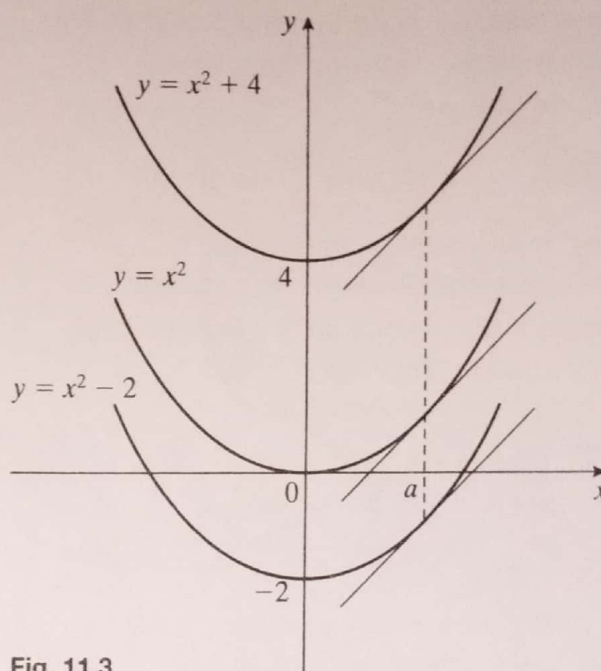


Fig. 11.3

Notice from Fig. 11.3 that for any given value of x , such as $x = a$, the gradients of the three curves are the same. For example, since $\frac{dy}{dx} = 2x$ for each, then if $a = 2$ the gradient is 4 for each curve.

Therefore, if $\frac{dy}{dx} = 4$, then $y = 4x + c$

and if $\frac{dy}{dx} = \frac{1}{2}$, then $y = \frac{1}{2}x + c$

and if $\frac{dy}{dx} = 2x$, then $y = x^2 + c$.

11.2 Integration

The process of finding the expression for y in terms of x when given the gradient function is called **integration**. Integration is the **reverse of differentiation**. $x^2 + c$ is called the **integral** of $2x$ with respect to x and c is called the **arbitrary constant**.

The equations $\frac{dy}{dx} = \frac{1}{2}$ and $\frac{dy}{dx} = 2x$ are called **differential equations** and $y = \frac{1}{2}x + c$, and $y = x^2 + c$, respectively, are the **general solutions**.